Finite structure in a min–max quadratic sign problem

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August 2026

Abstract

For a symmetric zero-diagonal sign matrix, let $M(A)$ be the largest absolute value of its quadratic form on the Boolean cube, and put $F(n) = \min_A M(A)$. We give an auditable account of the exact values through order fifteen, including explicit computational trust boundaries. Direct enumeration stops at order twelve; orders thirteen through fifteen follow from a one-vertex lift together with a parity law, and give $F(15) = 27$. We also prove an exact energy-weighted covering formula for one-vertex extension, record finite non-heredity of optimizers, and give a sharp completion theorem for one fixed oriented-Hadamard frame which implies $F(16) \le 30$. The asymptotic question whether $F(n)/n^{3/2}$ converges remains open.

Status. This is a focused manuscript draft. Every unconditional analytic statement is proved below. Claims labelled computer-assisted depend on the stated enumeration boundary. No result here proves convergence or nonconvergence.

1 Problem and elementary structure

Let $A = (a_{ij})$ range over symmetric zero-diagonal matrices with $a_{ij} \in \{-1, 1\}$. Define

$$Q_A(x) = \sum_{1\le i<j\le n} a_{ij}x_ix_j, \qquad M(A) = \max_{x\in\{-1,1\}^n} |Q_A(x)|, \qquad F(n) = \min_A M(A).$$

Switching by $z \in \{-1, 1\}^n$ sends A to D_zAD_z and leaves $M(A)$ unchanged. Vertex permutation and global negation also preserve $M(A)$.

Lemma 1.1 (Parity law)**.** *Let A be a symmetric zero-diagonal matrix of order n with off-diagonal entries in $\{-1, 1\}$. Then*

$$Q_A(x) \equiv \binom{n}{2} \pmod 2 \qquad \text{for every } x \in \{-1,1\}^n,$$

and consequently $M(A) \equiv \binom{n}{2} \pmod 2$ for every such A and

$$F(n) \equiv \binom{n}{2} \pmod 2.$$

Proof. $Q_A(x)$ is a sum of the $\binom{n}{2}$ terms $a_{ij}x_ix_j$, each of which lies in $\{-1, 1\}$. A sum of N terms from $\{-1, 1\}$ equals $N - 2k$, where k is the number of terms equal to -1, so it has the parity of N. Hence every value of Q_A on $\{-1, 1\}^n$ is congruent to $\binom{n}{2}$ modulo 2, and so is $|Q_A(x)|$. Taking the maximum over x preserves the residue, giving $M(A) \equiv \binom{n}{2}$, and taking the minimum over A preserves it again. $\square$

Lemma 1.2 (Puncturing)**.** *Let A be a signing of order $n+1$ and let B be its principal submatrix on any n of the vertices. Then $M(B) \le M(A)$. Consequently, for every $n \ge 2$,*

$$F(n) \le F(n+1) \le F(n) + n,$$

and, in combination with Lemma 1.1,

$$F(n+1) \ \ge \ F(n) + (n \bmod 2). \tag{1}$$

Proof. Relabel so that B occupies the first n vertices and write $v \in \{-1,1\}^n$ for the signs incident with the deleted vertex. For $x \in \{-1,1\}^n$ and $t \in \{-1,1\}$,

$$Q_A(x,t) = Q_B(x) + t\,(v \cdot x). \tag{2}$$

Averaging (2) over $t = \pm 1$ gives $Q_B(x) = \frac{1}{2}\big(Q_A(x,1) + Q_A(x,-1)\big)$, so $|Q_B(x)| \le M(A)$ for every x and therefore $M(B) \le M(A)$. Minimising over A at order $n+1$ gives $F(n) \le F(n+1)$. For the upper bound, extend an optimal order-n signing by an arbitrary v: by (2) every energy moves by $|v \cdot x| \le n$, so the maximum grows by at most n. Finally $\binom{n+1}{2} - \binom{n}{2} = n$, so for odd n the two orders have opposite parities and Lemma 1.1 forbids $F(n+1) = F(n)$; with $F(n+1) \ge F(n)$ this is (1). $\square$

Remark 1.3. The parity law is a check on the whole table rather than on one entry. Since $\binom{n}{2}$ is even for $n \equiv 0, 1 \pmod 4$ and odd for $n \equiv 2, 3 \pmod 4$, the predicted parities for $n = 2, \ldots, 15$ are

odd, odd, even, even, odd, odd, even, even, odd, odd, even, even, odd, odd,

and the fourteen known values $1, 3, 4, 4, 5, 9, 10, 12, 13, 17, 18, 20, 21, 27$ agree in every position. Three of those fourteen values, at $n = 13$, $n = 14$ and $n = 15$, are established below using the law rather than merely checked against it.

There is also a useful exact coding formulation. Put $E_n = \binom{n}{2}$ and

$$D_n = \{(x_i x_j)_{i<j} : x \in \{-1,1\}^n\} \cup \{-(x_i x_j)_{i<j} : x \in \{-1,1\}^n\}.$$

If $\rho(D_n)$ denotes its covering radius in the Hamming cube, then

$$F(n) = E_n - 2\rho(D_n). \tag{3}$$

Indeed, correlation with a sign word is $E_n - 2$ times Hamming distance, and adjoining the negatives converts absolute correlation to nearest-codeword distance. This identity is exact but does not by itself control the second-order deficit of size $n^{3/2}$.

2 Exact values through order fifteen

Theorem 2.1 (Exact finite sequence)**.** *The exact values from order two through order fifteen are*

n	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$F(n)$	1	3	4	4	5	9	10	12	13	17	18	20	21	27.

Every lower bound through order twelve is computer-assisted, with the enumeration boundary recorded in Section 2.3. Orders thirteen and fourteen use no enumeration beyond order twelve: they follow from $F(12) = 18$ by Proposition 2.5. Order fifteen uses no enumeration beyond order twelve either, but does require the lift of Lemma 2.3 to be iterated three times; this is Proposition 2.7.

Certificate architecture. For orders at most ten, switching makes every edge incident with a chosen root positive. The remaining signing is an ordinary graph on $n-1$ vertices. Isomorph-free generation, exact Gray-code energy evaluation, and independent decoding reproduce

$$1, 3, 4, 4, 5, 9, 10, 12, 13.$$

All energy and threshold decisions use integers.

For orders eleven and twelve the same enumeration is run without any pruning filter. At order eleven it decodes all 12,005,168 unlabeled graphs on ten residual vertices and returns minimum seventeen. At order twelve it decodes all 1,018,997,864 unlabeled graphs on eleven residual vertices, returns minimum eighteen, and at the same time collects every class whose maximum is at most eighteen. There are exactly two. Every energy comparison is an exact integer comparison, and no class is discarded by a degree or energy heuristic before its maximum is computed.

Orders thirteen and fourteen require no further enumeration. They follow from $F(12) = 18$, from the two collected classes, and from Lemmas 1.1, 1.2 and 2.3; this is Proposition 2.5.

The exhaustive lower certificates through order twelve trust the completeness of nauty's isomorph-free generation. They do not infer infeasibility from a solver timeout, and they apply no pruning filter whose soundness would have to be audited separately. The explicit upper witnesses, their complete energy scans, and the one-vertex lift of Lemma 2.3 are reproducible without that trust boundary. $\square$

Remark 2.2. The order-twelve scan finds two *rooted records*, that is two classes under the switching-permutation group with a distinguished root, not a theorem that there are exactly two unrooted optimal switching classes at order twelve. Proposition 2.5 only needs the rooted list to be complete, which is what the enumeration provides. Likewise, the certificate proves that the order-fourteen conference matrix is optimal but does not prove uniqueness.

2.1 The one-vertex lift

For an order-n signing B and $v \in \{-1, 1\}^n$, let $B \oplus v$ denote the order-$(n+1)$ signing whose leading principal block is B and whose last row and column are v, and put

$$E(B) = \min_{v \in \{-1,1\}^n} M(B \oplus v). \tag{4}$$

Lemma 2.3 (One-vertex lift)**.** *For every order-n signing B and every $v \in \{-1, 1\}^n$,*

$$M(B \oplus v) = \max_{x \in \{-1,1\}^n} \big(|Q_B(x)| + |v \cdot x|\big). \tag{5}$$

Let T be an integer and let $\mathcal{B}_n(T)$ contain one representative of each switching-permutation class of order-n signings with $M \le T$. Then, with the convention $\min \emptyset = +\infty$*,*

$$F(n+1) \le T \quad \Longleftrightarrow \quad \min_{B \in \mathcal{B}_n(T)} E(B) \le T. \tag{6}$$

Proof. Fix $x \in \{-1, 1\}^n$. By (2) the two choices of the new spin give the values $Q_B(x) \pm (v \cdot x)$, and the larger of their absolute values is $|Q_B(x)| + |v \cdot x|$. Maximising over x gives (5).

The quantity $E(B)$ is invariant under the symmetry group: if $B' = PD_zBD_zP^{\mathsf{T}}$ for a signing z and a permutation matrix P, then $B' \oplus (PD_zv) = \widetilde{P}\widetilde{D}(B \oplus v)\widetilde{D}\widetilde{P}^{\mathsf{T}}$ with $\widetilde{D} = \operatorname{diag}(D_z, 1)$ and $\widetilde{P} = \operatorname{diag}(P, 1)$, so $v \mapsto PD_zv$ permutes $\{-1, 1\}^n$ and leaves the multiset $\{M(B \oplus v) : v \in \{-1, 1\}^n\}$ unchanged. Hence $E(B') = E(B)$.

Suppose $F(n+1) \le T$ and let A be an order-$(n+1)$ signing with $M(A) \le T$. Delete its last vertex: by Lemma 1.2 the remaining block B_0 has $M(B_0) \le T$, and $A = B_0 \oplus v$ where v carries the deleted incident signs. So $\mathcal{B}_n(T)$ is nonempty and its representative B of the class of B_0 satisfies $E(B) = E(B_0) \le M(A) \le T$. Conversely, any B with $E(B) \le T$ exhibits an order-$(n+1)$ signing of maximum at most T, so $F(n+1) \le T$. $\square$

Remark 2.4 (Restriction to minimisers, and cost). Take $T = F(n)$ in (6). Any order-$(n+1)$ signing with $M \le F(n)$ restricts, on each of its $n+1$ principal n-subsets, to an order-n signing with $M \le F(n)$, that is to a *minimiser* at order n. So deciding whether $F(n+1) = F(n)$ needs only the order-n minimisers, however many non-minimising classes there are. Each one costs a single dense product: with $X \in \{-1,1\}^{2^n \times n}$ listing all sign vectors and $q \in \mathbb{Z}^{2^n}$ the vector of energies $Q_B(x)$, the entry of $q\mathbf{1}^\mathsf{T} + XX^\mathsf{T}$ in row x and column v is $Q_B(x) + v \cdot x$, so

$$M(B \oplus v) = \max_x \big|q\mathbf{1}^\mathsf{T} + XX^\mathsf{T}\big|_{xv}, \qquad E(B) = \min_v \max_x \big|q\mathbf{1}^\mathsf{T} + XX^\mathsf{T}\big|_{xv}.$$

The absolute value absorbs the pairing over the new spin because x and $-x$ both occur as rows. For $n = 12$ this is one 4096×4096 product per class. Section 3 refines (5) into an exact covering formula for $E(B)$; only (6) is needed here.

Proposition 2.5 (Orders thirteen and fourteen)**.** *$F(13) = 20$ and $F(14) = 21$. Both follow from $F(12) = 18$ together with the complete order-twelve minimiser list, Lemmas 1.1, 1.2 and 2.3, and two explicit witnesses. No enumeration beyond order twelve is used, and order fourteen uses no search at all.*

Proof. The order-twelve enumeration returns $F(12) = 18$ and the complete list $\mathcal{B}_{12}(18)$, which has exactly two elements. In the rooted encoding of Section 2.3 their graph6 codes are `JCpdUg{[dM?` and `JCpVdXyxpz?`.

Lower bound at thirteen. Apply Lemma 2.3 with $n = 12$ and $T = 18$. Evaluating (4) for both classes over all $2^{12} = 4096$ sign vectors v gives $E(B) = 24$ for each, so

$$\min_{B \in \mathcal{B}_{12}(18)} E(B) = 24 > 18,$$

and (6) gives $F(13) > 18$. By Lemma 1.1, $F(13) \equiv \binom{13}{2} = 78 \equiv 0 \pmod 2$, so $F(13) \ne 19$ and $F(13) \ge 20$.

Upper bound at thirteen. Let C be the order-fourteen Paley conference matrix, so C is symmetric with zero diagonal, off-diagonal entries in $\{-1,1\}$, and $C^2 = 13I$. Deleting any one vertex leaves an order-thirteen signing; the complete integer energy scan over all 2^{13} sign vectors gives maximum 20, for each of the fourteen choices of deleted vertex. Hence $F(13) \le 20$ and therefore $F(13) = 20$.

Order fourteen. Lemma 1.2 gives $F(14) \ge F(13) = 20$, and Lemma 1.1 gives $F(14) \equiv \binom{14}{2} = 91 \equiv 1 \pmod 2$, so $F(14) \ne 20$ and $F(14) \ge 21$. Equivalently, (1) with $n = 13$ odd gives $F(14) \ge F(13) + 1 = 21$ at once. The complete energy scan of C itself gives $M(C) = 21$. Hence $F(14) = 21$. $\square$

Remark 2.6 (Uniqueness of the minimisers). The enumerations record rooted representatives, and one switching class contributes one representative per orbit of its automorphism group on the root, so the raw counts overstate the number of classes. Canonicalising every rooting with nauty's `labelg` collapses them: at order twelve the two records `JCpdUg{[dM?` and `JCpVdXyxpz?` are a single class, and at orders thirteen and fourteen there is a single record with a single rooting. Hence the minimiser is unique up to switching and permutation at orders twelve, thirteen and fourteen. The unique minimiser at order fourteen is the Paley conference matrix C, and the unique minimiser at order thirteen is C with one vertex deleted, which is consistent with the fact that all fourteen deletions give the same maximum. At order ten the four records form two classes, so $F(10) = 13$ has two minimisers.

2.2 Order fifteen

Direct enumeration is unavailable at order fifteen: it would require scoring the 50,502,031,367,952 graphs on thirteen vertices. Iterating the lift instead keeps every level inside a complete set.

Proposition 2.7 (Order fifteen)**.** *$F(15) = 27$.*

Proof. Write $\mathcal{B}_n(T)$ as in Lemma 2.3. By Lemma 1.2, deleting a vertex from a signing of order $n+1$ with $M \le T$ leaves one of order n with $M \le T$, so $\mathcal{B}_{n+1}(T)$ is contained in the set of one-vertex extensions of $\mathcal{B}_n(T)$; and by Lemma 1.1 the threshold at each order may be lowered to the largest integer of the correct parity. For the target $T = 25$ at order fifteen this gives thresholds 24 at orders twelve and thirteen and 25 at orders fourteen and fifteen.

Seeding with the complete order-twelve enumeration and extending three times, with isomorph-free reduction by `labelg` at each level, gives

$$|\mathcal{B}_{12}(24)| = 82{,}502{,}142, \quad |\mathcal{B}_{13}(24)| = 282{,}202{,}131, \quad |\mathcal{B}_{14}(25)| = 1{,}313{,}164,$$

and then $\mathcal{B}_{15}(25) = \emptyset$. By (6) no signing of order fifteen has $M \le 25$, so $F(15) \ge 26$, and Lemma 1.1 with $\binom{15}{2} = 105$ odd gives $F(15) \ge 27$.

For the upper bound, extend the order-fourteen conference matrix C by each of the 2^{14} sign rows and scan every energy exactly. The multiset of maxima is $27, 29, 31, 33, 35$ with multiplicities $28, 1820, 4732, 1456, 156$, so $E(C) = 27$ and $F(15) \le 27$. Hence $F(15) = 27$. $\square$

Corollary 2.8 (Order sixteen)**.** *$F(16) \in \{28, 30\}$.*

Proof. Lemma 1.2 gives $F(16) \ge F(15) = 27$, and Lemma 1.1 with $\binom{16}{2} = 120$ even gives $F(16) \ge 28$. The pinned oriented-Hadamard completion of Section 5 gives $F(16) \le 30$. $\square$

Remark 2.9. Deciding between the two values needs $\mathcal{B}_{15}(27)$, the complete set of order-fifteen minimisers, which sits one threshold above the tower of Proposition 2.7. The level sizes at order fourteen for thresholds 21, 23 and 25 are 1, 4 and 1,313,164, so the next threshold is not a small step.

Remark 2.10 (Cost and independent recheck). The three extension levels are the only expensive step, and the last one is cheap: a sign row can be rejected as soon as it violates the threshold on any one energy, so screening against the most extreme energies first removes almost every candidate before the full scan. The final level was recomputed by a separately written routine that re-lifted all 1,313,164 records of $\mathcal{B}_{14}(25)$ and again found nothing at $M \le 25$. That routine was calibrated by raising the threshold to 27, where it recovers the witness above, so its null result is not vacuous.

Remark 2.11 (Independent check of the lift). The same procedure reproduces a value that was obtained independently by exhaustion. At order ten the enumeration returns $F(10) = 13$ and exactly four rooted records attaining it, with graph6 codes `HCRbczQ`, `HCpdehU`, `HCrbdxz`, `HCZbeyz`; by Remark 2.6 these are two switching classes, paired as {`HCRbczQ`, `HCrbdxz`} and {`HCpdehU`, `HCZbeyz`}. Their extension values (4) are $17, 19, 17, 19$. So (6) with $T = 13$ gives $F(11) > 13$, while the minimising v exhibits an order-eleven signing of maximum seventeen; with Lemma 1.1 this gives $15 \le F(11) \le 17$. The lift alone does not close the gap, but the upper witness it produces is exactly the value $F(11) = 17$ that the unpruned order-eleven enumeration returns by a different route, which is the check intended here. The spread $17, 19$ across records of equal maximum is the same failure of scalar state that Section 4 records at order nine: the minimiser property at order n does not determine the extension value at order $n + 1$.

2.3 Computational provenance

Every exhaustive statement above comes from one enumeration, run once per order. Switching by a diagonal sign matrix makes all edges at a chosen root positive, so each switching class of order n has a representative whose last row and column are $+1$, and the rest of the matrix is the signed adjacency of an ordinary graph G on the remaining $n - 1$ vertices, with $a_{ij} = -1$ exactly on the edges of G. Since M is invariant under switching and relabelling,

$$F(n) = \min_G M(A(G)),$$

the minimum over non-isomorphic graphs G on $n-1$ vertices. Those graphs are produced by nauty's `geng`, decoded from graph6 in blocks, and scored by an exact integer evaluation of $Q_{A(G)}$ over all 2^{n-1} sign vectors with the root spin fixed to $+1$. Nothing is filtered: every generated graph has its maximum computed.

The three largest enumerations consumed

$$274{,}668, \qquad 12{,}005{,}168, \qquad 1{,}018{,}997{,}864$$

graphs, on 9, 10 and 11 vertices, settling $F(10) = 13$, $F(11) = 17$ and $F(12) = 18$. These three counts equal the published numbers of non-isomorphic graphs on 9, 10 and 11 vertices, sequence `A000088` of [2]; they were also recomputed here directly from Pólya's enumeration theorem, independently of `geng`, and agree. The generator's exit status and the absence of trailing bytes in its stream are asserted at the end of each run, so a truncated stream fails rather than reporting a spurious minimum. The order-twelve run also emits every class of maximum at most eighteen; it emits exactly two, and those two are the input to Proposition 2.5.

No graph on twelve or more vertices is generated anywhere in this chain. The order-twelve run takes about twenty minutes on one laptop core, order eleven about eight seconds, order ten under a second. The lift of Proposition 2.5 is two 4096×4096 products and takes about a fifth of a second. The order-thirteen and order-fourteen witnesses are single energy scans of a 14×14 integer matrix and its principal submatrices. The trust boundary is therefore exactly one item: the completeness of `geng` at eleven vertices, cross-checked against [2] and against Pólya's theorem.

3 The exact one-vertex state

For an order-n signing B, let $E(B)$ be the smallest maximum obtainable by adjoining one vertex with arbitrary incident sign vector. Write

$$\mathbb{P}_n = \{-1, 1\}^n / \{x \sim -x\}, \qquad d_\pm([u], [v]) = \min\{d_H(u, v), d_H(u, -v)\}.$$

Put $M = M(B)$ and define

$$w_B([x]) = \frac{M - |Q_B(x)|}{2}, \qquad \rho_{\mathrm{w}}(B) = \max_{[b] \in \mathbb{P}_n} \min_{[x] \in \mathbb{P}_n} \big(d_\pm([b], [x]) + w_B([x])\big).$$

The weights are nonnegative integers because all order-n energies have the same parity.

Theorem 3.1 (Weighted Bellman identity)**.** *For every order-n signing B,*

$$E(B) = M(B) + n - 2\rho_{\mathrm{w}}(B). \tag{7}$$

Equivalently, with $\delta_{\mathrm{w}}(B) = \lfloor n/2 \rfloor - \rho_{\mathrm{w}}(B)$,

$$F(n+1) = \min_B \bigl(M(B) + (n \bmod 2) + 2\delta_{\mathrm{w}}(B) \bigr). \tag{8}$$

Only configurations satisfying $|Q_B(x)| \geq M(B) - 2\lfloor n/2 \rfloor$ can affect the inner covering minimum.

Proof. Let $b \in \{-1,1\}^n$ be the incident signs and t the new spin. Pairing the two values of t gives

$$\max_{t \in \{-1,1\}} |Q_B(x) + t\, b \cdot x| = |Q_B(x)| + |b \cdot x|.$$

Since $|b \cdot x| = n - 2d_{\pm}([b],[x])$, the maximum over old spins for fixed b equals

$$M + n - 2 \min_{[x] \in \mathbb{P}_n} \bigl(w_B([x]) + d_{\pm}([b],[x]) \bigr).$$

Minimizing over $[b]$ proves (7). Minimizing over B and using $n - 2\lfloor n/2 \rfloor = n \bmod 2$ proves (8). If $w_B([x]) > \lfloor n/2 \rfloor$, that state loses to an exact maximizer at projective distance at most $\lfloor n/2 \rfloor$, proving the final claim. $\square$

The formula explains why exact optimizers need not be valid predecessors. It also shows why a scalar energy is not a closed state: the spatial placement of every energy layer in the projective cube matters.

4 Finite optimizer non-heredity

For fixed internal blocks A and B, define

$$J(A,B) = \min_{C \in \{-1,1\}^{s \times k}} M \begin{pmatrix} A & C \\ C^{\mathsf{T}} & B \end{pmatrix}.$$

Proposition 4.1 (A complete $2+8$ obstruction)**.** *Among optimal blocks of orders two and eight,*

$$\min_{M(A)=F(2),\, M(B)=F(8)} J(A,B) = 15 > F(10) = 13.$$

Consequently no optimal order-ten signing contains an optimal order-eight principal submatrix. The smallest internal maximum among order-eight blocks admitting a $2+8$ completion of maximum thirteen is twelve.

This is a finite exhaustive proposition. The verifier enumerates every relevant switching-permutation class, every rectangular cross block modulo the valid symmetries, and every projective full spin. It establishes an "internal sacrifice" phenomenon: minimizing the blocks separately can make their joint optimum worse.

There is a second exact failure of scalar state. Two optimal order-nine classes have the same complete projective absolute-energy histogram

$$\#\{|Q_B| = 0, 4, 8, 12\} = (60, 111, 60, 25),$$

so all their scalar partition functions agree at every temperature, but their minimum one-vertex extension values are thirteen and fifteen. Their weighted covering geometry, not their energy multiset, distinguishes them.

5 A sharp oriented-Hadamard completion

Let

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \end{pmatrix}, \qquad (a_{01}, a_{02}, a_{03}, a_{12}, a_{13}, a_{23}) = (1, 1, 1, -1, 1, -1).$$

For arbitrary symmetric zero-diagonal order-four signings D_0, D_1, D_2, D_3, let $L(D_0, D_1, D_2, D_3)$ have diagonal blocks D_i, upper cross block ij equal to $a_{ij}H$, and transpose blocks below.

Theorem 5.1 (Sharp completion of the pinned frame)**.**

$$\min_{D_0, D_1, D_2, D_3} M(L(D_0, D_1, D_2, D_3)) = 30. \tag{9}$$

In particular, $F(16) \le 30$.

Proof. Put

$$u_0 = (1, 1, 1, 1), \quad u_1 = (1, 1, 1, -1), \quad u_2 = (1, 1, -1, 1), \quad u_3 = (1, 1, -1, -1).$$

Direct multiplication gives the following cross-only energies:

	(x_0, x_1, x_2, x_3)	Q_{cross}
1+	$(u_0, u_0, -u_0, u_1)$	28
1−	$(u_0, -u_3, -u_0, -u_1)$	−28
2+	$(u_3, u_3, -u_3, u_2)$	28
2−	$(u_3, -u_2, -u_3, -u_2)$	−28
3+	$(u_3, u_2, -u_3, u_2)$	28
3−	$(u_3, -u_0, -u_3, -u_2)$	−28.

Within each pair the projective spins on clouds zero, two, and three agree, so their internal contributions cancel in the comparison. If the full maximum were at most twenty-eight, the three comparisons would force

$$q_{D_1}(u_0) \le q_{D_1}(u_3) \le q_{D_1}(u_2) \le q_{D_1}(u_0).$$

All three values would be equal. But

$$q_{D_1}(u_0) - q_{D_1}(u_2) = 2(d_{13} + d_{23} + d_{34}) \ne 0,$$

because a sum of three signs cannot vanish. Order-sixteen energies are even, so the restricted minimum is at least thirty.

For attainment, list the six internal edges in order $01, 02, 03, 12, 13, 23$ and take

$$P = (1, 1, 1, -1, 1, -1), \qquad R = (-1, -1, 1, 1, 1, -1), \qquad (D_0, D_1, D_2, D_3) = (P, R, P, R).$$

An exhaustive exact scan of all 2^{15} projective spins gives maximum thirty. Independent direct and blockwise implementations reproduce the complete energy histogram. □

Both P and R have maximum four and are switching-permutation equivalent. Nevertheless their ordered response vectors to this frame are

$$(2, 0, 4, -2, 0, 2, -2, -4), \qquad (0, -2, 2, -4, -2, 0, 4, 2).$$

Using one literal common internal block in all four clouds has minimum thirty-eight. Thus neither $M(D)$ nor the unframed switching class is a closed substitution state; orientation relative to the cross frame is essential. The theorem is restricted to one frame. It neither proves $F(16) = 30$ nor supplies an iterable amplification law.

6 Reproducibility and scope

The repository contains source-built verifiers for all statements above. The principal commands are:

```
python3 verification/verify_F_exhaustive.py 10 11
python3 verification/lift.py verification/order12_minimisers.g6 12 18
python3 verification/class_count.py verification/order12_minimisers.g6
python3 verification/research_cross_block_composition.py
python3 verification/verify_nonlinear_bellman.py
python3 verification/verify_framed_hadamard_lift_30.py
```

The first command reruns the unpruned enumeration at orders ten and eleven; adding `12` settles $F(12) = 18$ in about twenty minutes and is the only enumeration the table needs. The second returns 24, which is the content of Proposition 2.5, and the third exhibits the two order-twelve records as one switching class. Order fifteen is the expensive step,

```
PASSES=8 JOBS=16 bash verification/tower25.sh
```

needing nauty, about fifteen gigabytes of scratch, and fifteen to twenty hours on sixteen cores; its final level can be rechecked alone in about twenty minutes with `verification/full_final_check.py`. Exact commands, pinned counts, expected outputs, corruption controls, and the producer trust boundary are recorded in `verification/README.md`.

These finite results identify optimizer-sensitive composition as the main structural difficulty. They do not determine an asymptotic constant and do not constitute an answer to the MathOverflow limit question. The companion manuscript `composition_framework.tex` isolates the remaining cross-order theorem.

References

[1] P. Ivanisvili, *Min-max of a quadratic form of plus-minus ones*, MathOverflow question 413935, 2022.

[2] OEIS Foundation Inc., *Sequence A000088: number of graphs on n unlabeled nodes*, The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A000088`.

[3] B. D. McKay and A. Piperno, *Practical graph isomorphism, II*, Journal of Symbolic Computation 60 (2014), 94–112.